

Pushing the boundaries

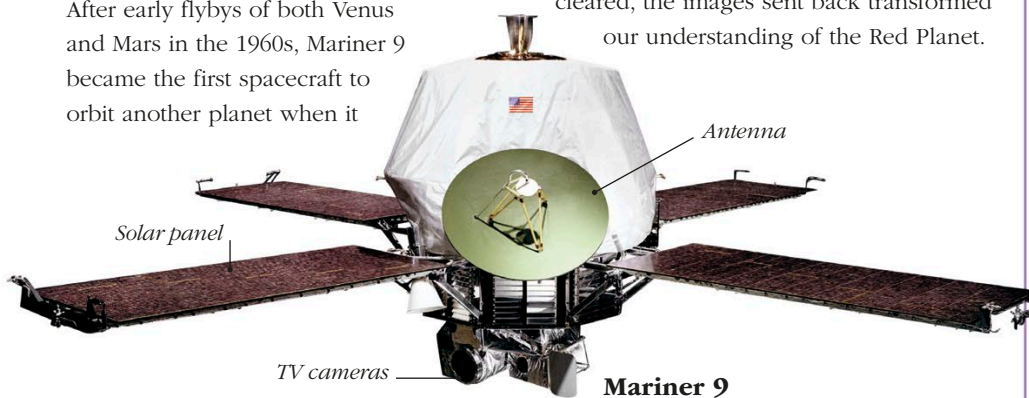
Since the late 1950s, dozens of robot probes have ventured to all parts of the solar system. Early missions concentrated on our moon in preparation for the manned Apollo landings. Our first survey of the major planets was completed by the late 1980s, and since then probes have become ever more complex and ambitious.

Orbiting Mars

- **What?** Mariner 9
- **Who?** NASA
- **Where and when?** US, launched 1971

After early flybys of both Venus and Mars in the 1960s, Mariner 9 became the first spacecraft to orbit another planet when it

arrived at Mars in November 1971. When the probe arrived, a planet-wide dust storm was in progress, but after that cleared, the images sent back transformed our understanding of the Red Planet.



Asteroid explorers

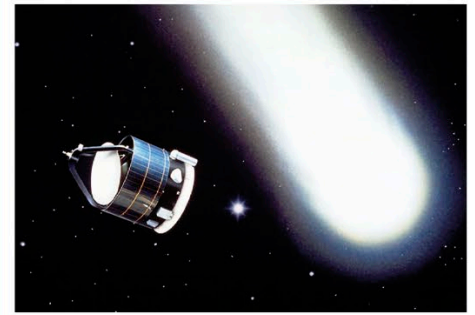
- **What?** NEAR Shoemaker
- **Who?** NASA
- **Where and when?** US, launched 1996

After several early flybys of small asteroids, NASA's Near Earth Asteroid Rendezvous Shoemaker (NEAR Shoemaker) probe spent a year orbiting the near-Earth asteroid Eros in 2000–2001, before eventually touching down on the asteroid's surface. More recently, a NASA mission called Dawn has investigated the solar system's two largest asteroids, Ceres and Vesta.



Illustration of NEAR Shoemaker orbiting Eros

Illustration of Giotto and Halley's Comet



Studying Halley's Comet

- **What?** Giotto
- **Who?** ESA
- **Where and when?** Europe, launched 1985

In the mid-1980s, space agencies launched probes to intercept Halley's Comet as it flew past the sun for the first time in 76 years. Russia, Japan, and Europe all took part. Launched in French Guiana, the European Space Agency's Giotto probe flew within 370 miles (596 km) of the comet's icy nucleus.

WOW!

Currently traveling through space at a distance of 13 billion miles (20.9 billion km) from Earth, Voyager 1 is the most distant man-made object ever launched.